

# **PURPLE COMPANY – E-COMMERCE ORDER MANAGEMENT**

## **DBMS – ORDER AND ORDER\_ITEM MANAGEMENT**

### **1. DATABASE DESIGN**

#### **1.1 ORDER TABLE**

| Attribute    | Description     |
|--------------|-----------------|
| Order_ID     | Unique Order ID |
| Customer_ID  | Customer ID     |
| Order_Date   | Date of order   |
| Total_Amount | Total amount    |
| Order_Status | Order status    |

#### **1.2 ORDER\_ITEM TABLE**

| Attribute     | Description          |
|---------------|----------------------|
| Order_Item_ID | Unique Order Item ID |
| Order_ID      | Order ID             |
| Product_ID    | Product ID           |
| Quantity      | Quantity ordered     |
| Price         | Product price        |

### **2. RELATIONSHIP**

#### **ORDER → ORDER\_ITEM**

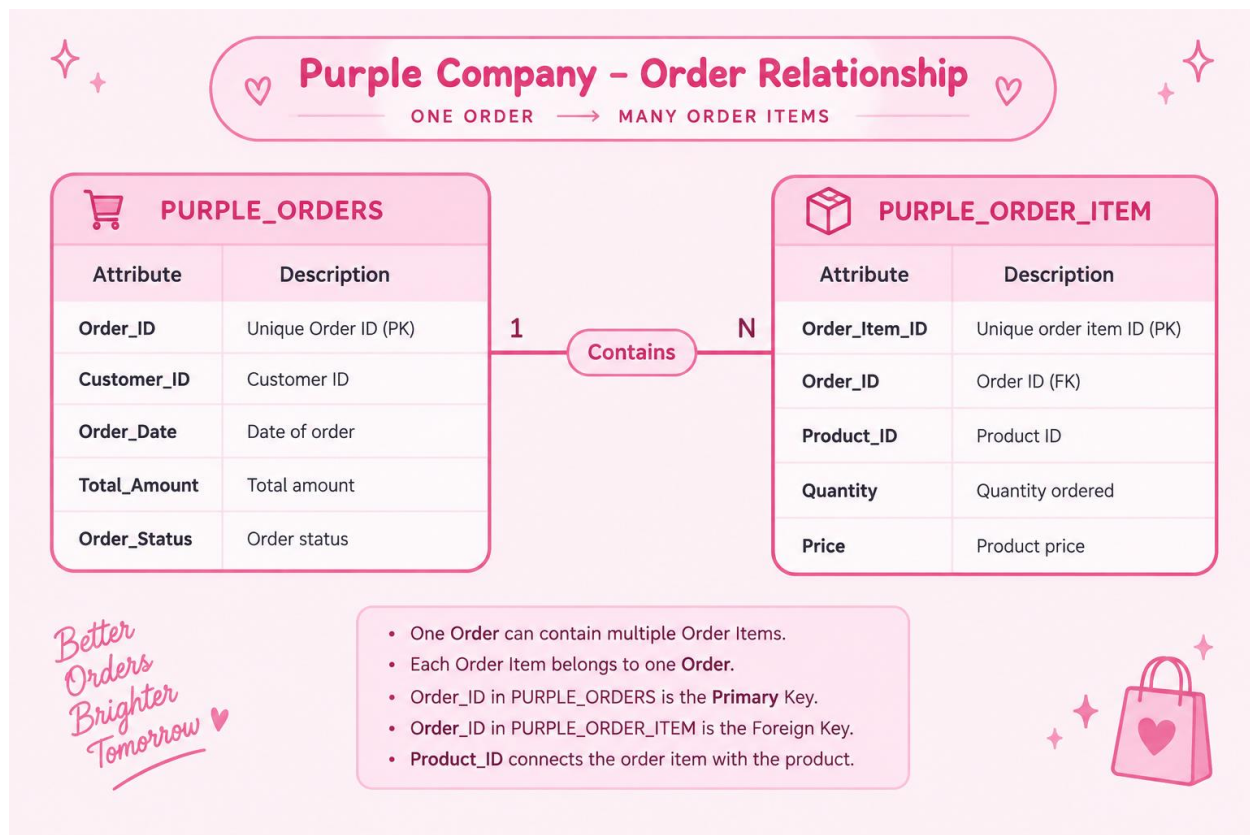
One Order can contain multiple Order Items.

Each Order Item belongs to one Order.

Order\_ID in the ORDER table is the Primary Key.

Order\_ID in the ORDER\_ITEM table is the Foreign Key.

Product\_ID connects the order item with the product.



### 3. TABLE CREATION

#### 3.1 ORDER TABLE

```
SQL> CREATE TABLE PURPLE_ORDERS (
2     Order_ID INT PRIMARY KEY,
3     Customer_ID INT,
4     Order_Date DATE,
5     Total_Amount DECIMAL(10,2),
6     Order_Status VARCHAR2(50)
7 );
```

Table created.

### 3.2 ORDER\_ITEM TABLE

```
SQL> CREATE TABLE PURPLE_ORDER_ITEM (  
2      Order_Item_ID INT PRIMARY KEY,  
3      Order_ID INT,  
4      Product_ID INT,  
5      Quantity INT,  
6      Price DECIMAL(10,2),  
7      FOREIGN KEY (Order_ID)  
8      REFERENCES PURPLE_ORDERS(Order_ID)  
9  );
```

### 4. ORDER TABLE DATA INSERTION

```
SQL>  
SQL> INSERT INTO PURPLE_ORDERS  
2  (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)  
3  VALUES  
4  (103, 3, '22-SEP-2026', 3200.00, 'Shipped');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO PURPLE_ORDERS  
2  (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)  
3  VALUES  
4  (104, 4, '23-SEP-2026', 1500.00, 'Pending');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO PURPLE_ORDERS  
2  (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)  
3  VALUES  
4  (105, 5, '23-SEP-2026', 4500.00, 'Delivered');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO PURPLE_ORDERS  
2  (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)  
3  VALUES  
4  (106, 6, '24-SEP-2026', 2750.00, 'Processing');  
  
1 row created.  
  
SQL>  
SQL> COMMIT;  
  
Commit complete.  
  
SQL> SELECT * FROM PURPLE_ORDERS;
```

## 4.1 DISPLAY ORDER TABLE DATA

```
SQL> SELECT * FROM PURPLE_ORDERS;
```

| ORDER_ID | CUSTOMER_ID | ORDER_DAT | TOTAL_AMOUNT | ORDER_STATUS |
|----------|-------------|-----------|--------------|--------------|
| 101      | 1           | 20-SEP-26 | 2500         | Delivered    |
| 102      | 2           | 21-SEP-26 | 1800         | Processing   |
| 103      | 3           | 22-SEP-26 | 3200         | Shipped      |
| 104      | 4           | 23-SEP-26 | 1500         | Pending      |
| 105      | 5           | 23-SEP-26 | 4500         | Delivered    |
| 106      | 6           | 24-SEP-26 | 2750         | Processing   |

## 5. ORDER\_ITEM TABLE DATA INSERTION

```
SQL> INSERT INTO PURPLE_ORDER_ITEM
2 (Order_Item_ID, Order_ID, Product_ID, Quantity, Price)
3 VALUES
4 (2, 102, 202, 1, 1800.00);
INSERT INTO PURPLE_ORDER_ITEM
*
ERROR at line 1:
ORA-00001: unique constraint (SYSTEM.SYS_C008461) violated

SQL>
SQL> INSERT INTO PURPLE_ORDER_ITEM
2 (Order_Item_ID, Order_ID, Product_ID, Quantity, Price)
3 VALUES
4 (3, 103, 203, 2, 1600.00);
1 row created.

SQL>
SQL> INSERT INTO PURPLE_ORDER_ITEM
2 (Order_Item_ID, Order_ID, Product_ID, Quantity, Price)
3 VALUES
4 (4, 104, 204, 1, 1500.00);
1 row created.

SQL>
SQL> INSERT INTO PURPLE_ORDER_ITEM
2 (Order_Item_ID, Order_ID, Product_ID, Quantity, Price)
3 VALUES
4 (5, 105, 205, 3, 1500.00);
1 row created.

SQL>
SQL> INSERT INTO PURPLE_ORDER_ITEM
2 (Order_Item_ID, Order_ID, Product_ID, Quantity, Price)
3 VALUES
4 (6, 106, 206, 2, 1375.00);
1 row created.
```

## 5.1 DISPLAY ORDER\_ITEM TABLE DATA

6 rows selected.

```
SQL> SELECT * FROM PURPLE_ORDER_ITEM;
```

| ORDER_ITEM_ID | ORDER_ID | PRODUCT_ID | QUANTITY | PRICE |
|---------------|----------|------------|----------|-------|
| 1             | 101      | 201        | 2        | 1000  |
| 2             | 102      | 202        | 1        | 1800  |
| 3             | 103      | 203        | 2        | 1600  |
| 4             | 104      | 204        | 1        | 1500  |
| 5             | 105      | 205        | 3        | 1500  |
| 6             | 106      | 206        | 2        | 1375  |

6 rows selected.

## 6. MODIFY ORDER TABLE DATA

```
SQL> UPDATE PURPLE_ORDERS  
2 SET Total_Amount = 5000  
3 WHERE Order_ID = 101;
```

1 row updated.

```
SQL> UPDATE PURPLE_ORDERS  
2 SET Order_Status = 'Delivered'  
3 WHERE Order_ID = 102;
```

1 row updated.

## 6.1 DISPLAY MODIFY ORDER TABLE DATA

Commit complete.

SQL>

SQL> SELECT \* FROM PURPLE\_ORDERS;

| ORDER_ID | CUSTOMER_ID | ORDER_DAT | TOTAL_AMOUNT |
|----------|-------------|-----------|--------------|
|----------|-------------|-----------|--------------|

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| ORDER_STATUS |
|--------------|
|--------------|

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|           |   |           |      |
|-----------|---|-----------|------|
| 101       | 1 | 20-SEP-26 | 5000 |
| Delivered |   |           |      |

|           |   |           |      |
|-----------|---|-----------|------|
| 102       | 2 | 21-SEP-26 | 1800 |
| Delivered |   |           |      |

|         |   |           |      |
|---------|---|-----------|------|
| 103     | 3 | 22-SEP-26 | 3200 |
| Shipped |   |           |      |

| ORDER_ID | CUSTOMER_ID | ORDER_DAT | TOTAL_AMOUNT |
|----------|-------------|-----------|--------------|
|----------|-------------|-----------|--------------|

-----

| ORDER_STATUS |
|--------------|
|--------------|

-----

|         |   |           |      |
|---------|---|-----------|------|
| 104     | 4 | 23-SEP-26 | 1500 |
| Pending |   |           |      |

|           |   |           |      |
|-----------|---|-----------|------|
| 105       | 5 | 23-SEP-26 | 4500 |
| Delivered |   |           |      |

|            |   |           |      |
|------------|---|-----------|------|
| 106        | 6 | 24-SEP-26 | 2750 |
| Processing |   |           |      |

6 rows selected.

## 7. MODIFY ORDER\_ITEM TABLE DATA

```
SQL> UPDATE PURPLE_ORDER_ITEM
  2  SET Quantity = 3
  3  WHERE Order_Item_ID = 201;

0 rows updated.

SQL> UPDATE PURPLE_ORDER_ITEM
  2  SET Price = 2500
  3  WHERE Order_Item_ID = 202;

0 rows updated.

SQL> COMMIT;
```

### 7.1 DISPLAY MODIFY ORDER\_ITEM TABLE DATA

```
SQL> SELECT * FROM PURPLE_ORDER_ITEM;
```

| ORDER_ITEM_ID | ORDER_ID | PRODUCT_ID | QUANTITY | PRICE |
|---------------|----------|------------|----------|-------|
| 1             | 101      | 201        | 2        | 1000  |
| 2             | 102      | 202        | 1        | 1800  |
| 3             | 103      | 203        | 2        | 1600  |
| 4             | 104      | 204        | 1        | 1500  |
| 5             | 105      | 205        | 3        | 1500  |
| 6             | 106      | 206        | 2        | 1375  |

```
6 rows selected.
```

## 8. CUSTOMER WISE ORDER SUMMARY

```
SQL> SELECT
  2     o.Customer_ID,
  3     COUNT(o.Order_ID) AS Total_Orders,
  4     SUM(o.Total_Amount) AS Total_Amount
  5 FROM PURPLE_ORDERS o
  6 GROUP BY o.Customer_ID
  7 ORDER BY o.Customer_ID;
```

| CUSTOMER_ID | TOTAL_ORDERS | TOTAL_AMOUNT |
|-------------|--------------|--------------|
| 1           | 1            | 5000         |
| 2           | 1            | 1800         |
| 3           | 1            | 3200         |
| 4           | 1            | 1500         |
| 5           | 1            | 4500         |
| 6           | 1            | 2750         |

6 rows selected.

```
SQL> SELECT
  2     o.Customer_ID,
  3     COUNT(DISTINCT o.Order_ID) AS Total_Orders,
  4     SUM(o.Total_Amount) AS Total_Order_Amount,
  5     SUM(oi.Quantity) AS Total_Quantity
  6 FROM PURPLE_ORDERS o
  7 JOIN PURPLE_ORDER_ITEM oi
  8 ON o.Order_ID = oi.Order_ID
  9 GROUP BY o.Customer_ID
 10 ORDER BY o.Customer_ID;
```

| CUSTOMER_ID | TOTAL_ORDERS | TOTAL_ORDER_AMOUNT | TOTAL_QUANTITY |
|-------------|--------------|--------------------|----------------|
| 1           | 1            | 5000               | 2              |
| 2           | 1            | 1800               | 1              |
| 3           | 1            | 3200               | 2              |
| 4           | 1            | 1500               | 1              |